

COVID19 in the United Kingdom

Elements of Modelling Real-World Infectious Diseases

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Modelling Real World Diseases

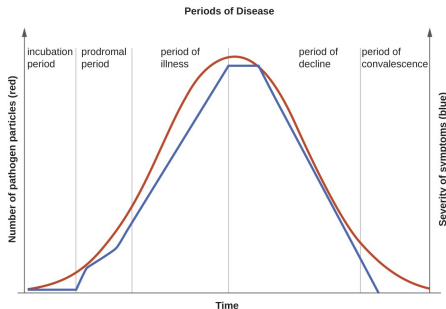
- SI, SIR models are appropriate for modelling some epidemics
- Some epidemics are more complicated
- What stages are there of the disease?
- Does each individual have equal chance of being infected?

Reinfection?

- SIR is good for disease which can only infect an individual once (e.g Chickenpox)
- SI is good for disease which an individual cannot recover from (e.g HIV)
- What if an individual can become infected more than once?
- SIS/SIRS is a model which can account for reinfections (e.g Influenza)

Infected AND Infectious?

- An individual is not infectious the moment they become infected
- A person becomes infectious when the level of the disease in the body is high enough
- For some diseases, it takes a while for an infected individual to become infectious
- SEIR models incorporate an "exposure" period



Stages of COVID-19

- There is thought to be an incubation period of 4-6 days
- Symptoms of COVID-19 vary between infected individual, but the disease is also very contagious.
- Individuals might be infectious but show no symptoms (Asymptomatic Stage)
- An individual might then become symptomatic, or remain asymptomatic.
- Re-infection of COVID-19 is very uncommon, so a removed state is appropriate.
- SEPIR?

Population Characteristics

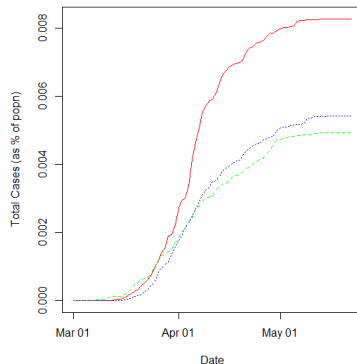
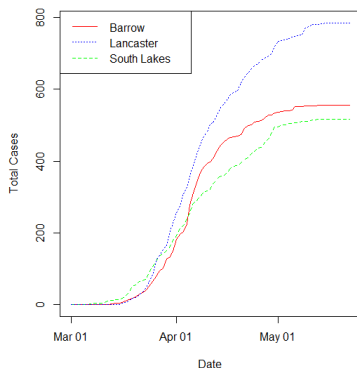
- Is it adequate to assume everyone is at equal risk to infection?
- Certain groups can be at higher/lower risk.
- Social Mixing

Population Characteristics

- Older individuals have been identified as higher risk
- Younger individuals are at lower risk
- Lockdown changes social mixing drastically
- Individuals make much more contact with people they live with.

Population Characteristics

- Spread of COVID-19 varies even at Local Authority level.
- Socio-economic factors may play a part in the spread.
- How compliant is the population with lockdown?
- How many key workers are in the population.



Thank You

Thank you for listening!
Feel free to ask any questions.